Timescale Robustness of the Adaptive Window in Viability-Constrained Memory Systems

VCSM Theory Series — Paper 8

2026

Abstract

Paper 7 established an empirical regime map for VCML adaptive memory substrates, identifying a bounded adaptive window in the (ν, κ) turnover–propagation plane. This paper asks: *what determines the location of that window?* We test the timescale-matching hypothesis—that the adaptive window occurs when $\tau_{\text{turnover}} \approx \tau_{\text{consolidation}}$—by independently varying the consolidation gate threshold (`SS`), consolidation learning rate (`FIELD_ALPHA`), and wave input rate (`WAVE_RATIO`). We find that the window's location in ν is *robustly stable* ($\nu^* \approx 0.001$) across a 4× range of each consolidation parameter, but varies when the wave input rate exceeds a perturbation threshold. A re-analysis of Paper 7 Experiment 3 confirms that the diffusion rate κ is similarly non-controlling: ν_{opt} is identical across all κ values tested. The main finding is a separation of concerns: *the regime boundary is set by input wave dynamics, not consolidation machinery.* Consolidation parameters control the amplitude of spatial structure within the window, not its location. We propose a consolidation amplitude formula relating `sg4` peak to `SS` and `WAVE_RATIO`, and discuss implications for designing adaptive memory substrates.

1 Introduction

Paper 7 (Regime Structure) mapped the (ν, κ) parameter space of a VCML substrate and identified four dynamical regimes: frozen (low ν), turnover-dominated (high ν), fragmented (low κ), and adaptive (intermediate ν, moderate κ). The adaptive window is bounded in ν between approximately 0.001 and 0.015, and is verified by three independent robustness experiments (classifier accuracy, copy-forward ablation, scaling).

A natural next question is: *what places the window at this location?* The timescale-matching hypothesis predicts:

$$\nu^* \approx \frac{c}{\tau_{\text{consolidation}}} \tag{1}$$

where $\tau_{\text{consolidation}}$ is set by `SS` (the calm-streak gate) and `FIELD_ALPHA` (the consolidation learning rate). If Eq. (1) holds, doubling `SS` should halve ν^*, and doubling `FIELD_ALPHA` should double ν^*.

Similarly, if the wave input rate `WAVE_RATIO` (WR) sets the *propagation timescale*, we would expect ν^* to scale proportionally with WR: faster input means the system can sustain higher turnover.

We test these predictions with three sweeps (Experiments 1–3), then analyze ν^* vs. κ from Paper 7 data (Experiment 4).

1.1 Background: VCSM Consolidation Gate

In VCSM, a site can only update its `fieldM` vector after accumulating `SS` consecutive unperturbed steps:

$$\mathrm{ok}[i] \;=\; [\texttt{streak}[i] \geq \texttt{SS}] \tag{2}$$

When ok[i] is true:

$$\texttt{fieldM}[i] \;\mathrel{+}=\; \texttt{FIELD_ALPHA} \cdot (\texttt{mid_mem}[i] - \texttt{fieldM}[i]) \tag{3}$$

The consolidation timescale is thus governed by two parameters: SS (gate threshold, steps) and FIELD_ALPHA (learning rate per step given calm). Slower FIELD_ALPHA means more calm steps are needed to accumulate a given field update; higher SS means a longer uninterrupted calm period is required before any update is possible.

The probability that a site achieves a calm streak $\geq$ `SS` at any given step depends on the wave perturbation density. For a half-grid site (left or right) and average WR active left-zone waves:

$$P_{\mathrm{calm}}(\texttt{SS}) \;\approx\; \left(1 - \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR}}\right)^{\texttt{SS}+\texttt{WAVE_DUR}-1} \tag{4}$$

This is the probability that no left-zone wave was launched in the preceding $\texttt{SS} + \texttt{WAVE_DUR} - 1$ steps (necessary for the site to have been calm for `SS` consecutive steps as of now).

2 Shared Protocol

All experiments use the Paper 7 Exp1 base configuration. Only the swept parameter varies; all others are fixed at reference values.

Parameter	Reference value
Grid	$W = 40$, $H = 20$, $\texttt{HALF} = 20$, $N_{\mathrm{act}} = 400$
`HS`	2
`KAPPA`	0.020
`WAVE_RATIO`	4.8
`WAVE_DUR`	15
`SS`	10
`FIELD_ALPHA`	0.16
`SEED_BETA`	0.25
`STEPS`	2000
`SHIFT`	1000
Seeds	5
ν sweep	$\{0.0001, 0.0003, 0.001, 0.002, 0.005, 0.010, 0.020, 0.040, 0.080\}$

For each parameter configuration, we report ν^* (argmax `sg4`) and `sg4`* (peak value, 5-seed mean). A failure of Eq. (1) would show constant ν^* across the sweep. A success would show $\nu^* \propto 1/\texttt{SS}$ or $\nu^* \propto \texttt{FIELD_ALPHA}$.

3 Experiment 1: Consolidation Gate Sweep

3.1 Protocol

$\texttt{SS} \in \{5, 10, 20, 40\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs. Expected: $\nu^* \propto 1/\texttt{SS}$ (inverse proportionality).

3.2 Results

Table 1: SS sweep: ν^* and $\texttt{sg4}^*$ by consolidation gate. sg4 profile shows 5-seed mean `sg4` at each ν value (left to right: $\nu = 0.0001, 0.0003, \ldots, 0.080$).

SS	ν^*	$\texttt{sg4}^*$	sg4 profile								
5	0.0010	887.8	120	576	**888**	734	126	18	5	1	0
10	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
20	0.0010	186.2	78	88	**186**	137	23	12	1	0	0
40	0.0020	23.5	0	0	0	**24**	5	0	0	0	0

Result. The location ν^* is unchanged at 0.001 for $\texttt{SS} = 5, 10, 20$ — a 4× range of gate values. At $\texttt{SS} = 40$, $\nu^* = 0.002$ (one grid point higher) and $\texttt{sg4}^* = 23.5$, near zero: the gate is so strict that almost no sites achieve 40 consecutive calm steps under the reference wave density.

The timescale-matching prediction ($\nu^* \propto 1/\texttt{SS}$) fails: the window's location is insensitive to SS. However, the *amplitude* $\texttt{sg4}^*$ drops steeply with SS: 888, 376, 186, 23.5 (approximately halving per SS doubling from 5 to 20, then collapsing at 40).

The amplitude drop is consistent with Eq. (4): doubling SS exponentially reduces P_{calm}, reducing the effective write rate.

4 Experiment 2: Consolidation Rate Sweep

4.1 Protocol

$\texttt{FIELD_ALPHA} \in \{0.04, 0.08, 0.16, 0.32\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs. Expected: $\nu^* \propto \texttt{FIELD_ALPHA}$ (faster learning tolerates more turnover).

4.2 Results

Table 2: FIELD_ALPHA sweep: ν^* and $\texttt{sg4}^*$.

`FIELD_ALPHA`	ν^*	$\texttt{sg4}^*$	sg4 profile								
0.04	0.0001	188.0	**188**	124	174	136	32	11	2	0	0
0.08	0.0010	263.1	213	191	**263**	226	52	19	4	1	0
0.16	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
0.32	0.0010	507.4	255	379	**507**	417	103	25	5	1	0

Result. $\nu^* = 0.001$ for $\texttt{FIELD_ALPHA} \in \{0.08, 0.16, 0.32\}$ (a 4× range). At $\texttt{FIELD_ALPHA} = 0.04$, ν^* shifts to 0.0001: very slow consolidation effectively freezes the system, requiring minimal turnover to avoid erasure.

The amplitude $\texttt{sg4}^*$ scales monotonically: 188, 263, 376, 507 (roughly proportional to FIELD_ALPHA in the 0.08–0.32 range). This confirms that FIELD_ALPHA controls the height of the adaptive peak but not its location, except at extreme values.

5 Experiment 3: Wave Input Rate Sweep

5.1 Protocol

`WAVE_RATIO` $\in \{1.2, 2.4, 4.8, 9.6\}$. All other parameters at reference. $4 \times 9 \times 5 = 180$ runs. Expected: $\nu^* \propto$ `WR` (more frequent input enables higher turnover).

5.2 Results

Table 3: WAVE_RATIO sweep: ν^* and $\mathtt{sg4}^*$.

WR	ν^*	$\mathtt{sg4}^*$	sg4 profile								
1.2	0.0010	120.4	32	104	**120**	68	16	7	1	0	0
2.4	0.0010	414.9	111	328	**415**	317	34	11	2	1	0
4.8	0.0010	376.3	223	273	**376**	328	80	24	4	1	0
9.6	0.0001	226.9	**227**	70	0	28	12	1	0	0	0

Result. $\nu^* = 0.001$ for `WR` $\in \{1.2, 2.4, 4.8\}$. At `WR` $= 9.6$, ν^* shifts to 0.0001 — *opposite* to the prediction. At WR=9.6 and $\nu = 0.001$, `sg4` $= 0$: high wave density combined with moderate turnover prevents any site from achieving `SS` $= 10$ consecutive calm steps. Only at very low turnover ($\nu = 0.0001$) does the system recover structure.

This confirms the over-perturbation collapse identified in Papers 7 and V93: excessive wave density suppresses consolidation, requiring near-zero turnover to compensate.

The amplitude $\mathtt{sg4}^*$ peaks at WR=2.4 (414.9) and is lower at WR=4.8 (376.3) and WR=1.2 (120.4), consistent with the V93 finding that WR$\sim 2\times$ is optimal at the $N = 400$ scale.

6 Experiment 4: Propagation Rate Analysis

6.1 Protocol

Re-analysis of Paper 7 Experiment 3 results. For each κ value, we identify $\nu_{\text{opt}} = \text{argmax}_\nu\, \mathtt{sg4}(\nu, \kappa)$ using the cached 5-seed means. This tests whether ν^* scales with κ (as a propagation timescale matching prediction would require).

6.2 Results

Table 4: ν_{opt} and $\mathtt{sg4}_{\text{opt}}$ by κ (Paper 7 Exp3 data). Collapse boundary is between $\nu = 0.005$ and $\nu = 0.015$ for all κ.

κ	ν_{opt}	$\mathtt{sg4}_{\text{opt}}$	sg4 profile (ν: 0.0003, 0.001, 0.005, 0.015, 0.050)				
0.000	0.0010	396.7	279	**397**	99	11	1
0.005	0.0010	383.1	271	**383**	88	9	1
0.020	0.0010	376.3	273	**376**	80	8	1
0.060	0.0010	375.1	273	**375**	68	6	1
0.150	0.0010	360.9	276	**361**	49	5	0

Result. $\nu_{\text{opt}} = 0.001$ for *all* κ values tested (0 to 0.15, a ∞-fold range). The turnover-dominated collapse boundary (between $\nu = 0.005$ and $\nu = 0.015$) is also unchanged across the full κ sweep. Diffusion κ shifts only the amplitude of `sg4`* (from 396.7 at $\kappa = 0$ to 360.9 at $\kappa = 0.15$), not the regime boundary location.

There is no evidence for $\nu_{\text{opt}} \propto \sqrt{\kappa}$ or any other κ-dependent scaling of the window location.

7 Discussion

7.1 The Regime Boundary Is Set by Input Dynamics

Across four experiments spanning the consolidation gate (SS), consolidation rate (FIELD_ALPHA), wave input rate (WR), and diffusion (κ), the location $\nu^* \approx 0.001$ is stable in all cases except extreme WR (9.6) or extreme FIELD_ALPHA (0.04). Within the practical operating range:

Parameter	Range tested	Effect on ν^*
`SS` (gate)	5–20	None ($\nu^* = 0.001$ throughout)
`FIELD_ALPHA` (rate)	0.08–0.32	None ($\nu^* = 0.001$ throughout)
`WR` (wave rate)	1.2–4.8	None ($\nu^* = 0.001$ throughout)
κ (diffusion)	0–0.15	None ($\nu^* = 0.001$ throughout)

The timescale-matching hypothesis ($\nu^* \propto 1/\texttt{SS}$) is *rejected* as a location law. The regime boundary is not placed by the consolidation timescale; it is placed by the copy-forward/erasure balance, which depends on the wave input protocol and grid geometry.

7.2 Separation of Concerns: Location vs. Amplitude

The experiments reveal a clean separation:

- **Location** (ν^*, regime boundary): set by wave input dynamics. Insensitive to consolidation parameters within a practical range. Shifts only under extreme over-perturbation (WR ≥ 9.6) or extreme under-writing (FIELD_ALPHA ≤ 0.04).
- **Amplitude** (`sg4`*, peak structure): strongly controlled by consolidation parameters. `sg4`* scales approximately proportionally to FIELD_ALPHA and drops exponentially with SS. κ shifts amplitude weakly (Exp4).

This has a practical implication: *to increase structure formation, tune FIELD_ALPHA and SS; to shift the regime boundary, change WR or WAVE_DUR.*

7.3 Amplitude Formula

The probability that a site achieves a calm streak $\geq$ `SS` (Eq. (4)) predicts the relative consolidation rate. For the reference WR = 4.8, WAVE_DUR = 15:

SS	P_{calm} (Eq. 4)	`sg4`* (observed)
5	0.047	888
10	0.015	376
20	0.0024	186
40	6×10^{-5}	23

The observed sg4* drops with SS qualitatively consistent with P_{calm}, though the exponential decay of P_{calm} overestimates the drop—likely because `sg4` saturates at high P_{calm} and does not grow linearly with the number of consolidation events.

7.4 Why $\nu^* \approx 0.001$ Is Universal

The stability of $\nu^* = 0.001$ across such diverse parameters suggests it reflects a *structural property of the wave–grid interaction*, not the consolidation machinery. At $\nu = 0.001$, each site dies on average once every 1000 steps. With `WAVE_DUR` = 15 and `WR` = 4.8, there are approximately $4.8 \times 2000/15 \approx 640$ total wave events across the run. Each site is involved in approximately half of them. This gives approximately 320 perturbation events per site per run—and with mean lifetime $1/\nu = 1000$, a site experiences about 0.32 deaths per lifetime per unit time.

At the copy-forward/erasure balance:

$$\text{copy-forward rate} \approx \text{erasure rate} \quad \Longrightarrow \quad \nu \approx \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR} \cdot N_{\text{act}}/4} \approx 0.0016 \tag{5}$$

This rough calculation places the balance near $\nu \approx 0.001$–0.002, consistent with the empirical ν^*. The consolidation parameters (SS, FIELD_ALPHA) do not enter this balance—they determine *how much* structure accumulates per successful copy-forward event, but not *when* the balance tips.

7.5 Implication for the Anomaly Region

Paper 7 Experiment 3 shows that at $(\nu = 0.005, \kappa = 0.060)$, `sg4` = 68 while adjacent cells (same ν, lower κ: `sg4` = 80; same κ, lower ν: `sg4` = 375) are substantially higher. This is not a grid artifact but the collapse onset: $\nu = 0.005$ sits near the right boundary of the adaptive window ($\nu_{\text{crit}} \approx 0.007$–$0.010$), and higher κ does not rescue structure because the boundary is determined by ν, not κ (Experiment 4). The cell at $(\nu = 0.005, \kappa = 0.060)$ is an accurate empirical location of the pre-collapse transition, not an anomaly.

7.6 Limitations

The wave input protocol is fixed (zone-based, 4 zones, structured alternation). The stability of ν^* may not hold for qualitatively different input statistics. The exponential P_{calm} formula is derived from an independence approximation; the true consolidation probability accounts for spatial correlations and streak-length distributions. Verification under continuous-time or variable-duration waves remains for future work.

8 Conclusion

We tested the timescale-matching hypothesis for the adaptive window location in VCML substrates. The hypothesis—that $\nu^* \propto 1/\texttt{SS}$ (consolidation gate)—is rejected: $\nu^* \approx 0.001$ is stable across a 4× range of SS, FIELD_ALPHA, and WR, and across the full κ sweep from 0 to 0.15.

The correct interpretation is a *separation of concerns*: the regime boundary is set by the copy-forward/erasure balance at the wave–grid level, and consolidation parameters control the amplitude of structure formed within the window. A rough balance equation ($\nu \approx \texttt{WR}/(2 \cdot \texttt{WAVE_DUR} \cdot N_{\text{act}}/4)$) correctly predicts the order of magnitude.

The adaptive window at $\nu \in [0.001, 0.010]$ is thus a property of the *input protocol*, not of the learning rule. To relocate the window, change the wave structure; to increase the quality of learning within it, adjust the consolidation machinery.

A Reproducibility

All experiments use only `numpy`. Experiment 4 uses cached JSON from Paper 7.

Script	Description	Runtime
`paper8_exp1_ss_sweep.py`	SS sweep (180 runs)	∼8 min
`paper8_exp2_alpha_sweep.py`	FIELD_ALPHA sweep (180 runs)	∼8 min
`paper8_exp3_wr_sweep.py`	WR sweep (180 runs)	∼8 min

Listing 1: Run all experiments.

```
cd papers/paper8_timescale_matching
py paper8_exp1_ss_sweep.py
py paper8_exp2_alpha_sweep.py
py paper8_exp3_wr_sweep.py
```